

Digital and Analog Electronics Portfolio

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Purpose:

The purpose of this portfolio is to demonstrate the practical implementation and investigation of a variety of digital and analog electronic circuits. All circuits were prototyped on a breadboard and supplied and measured with the Analog Discovery 2 by Digilent.

Instrumentation:

The Analog Discovery 2 works to supply both DC and waveform voltages. It also features two oscilloscope channels capable of measuring two voltages simultaneously.

Tools used:

- Digilent Analog Discovery 2
- LTspice Software
- Breadboard
- Electronic components

Bridge Rectifier, Filter and Capacitance Multiplier

Context:

Rectifiers are used to convert AC sources and signals into DC. The purpose of this project is to build and test a bridge rectifier with a few different kinds of filters to determine which works best to attenuate ripple and produce a flat DC voltage. The bridge rectifier works by passing an alternating current through a network of diodes. The positive end of the diode network supplies a positive, strobing voltage, eliminating the negative component of the input wave.

An AC source ($5V_{max}$) enters a network of diodes. A pulsating, positive voltage results, coming from the positive pole of the diode bridge. Figure 1 shows the input voltage (in blue) and the resulting pulsing voltage (in orange).

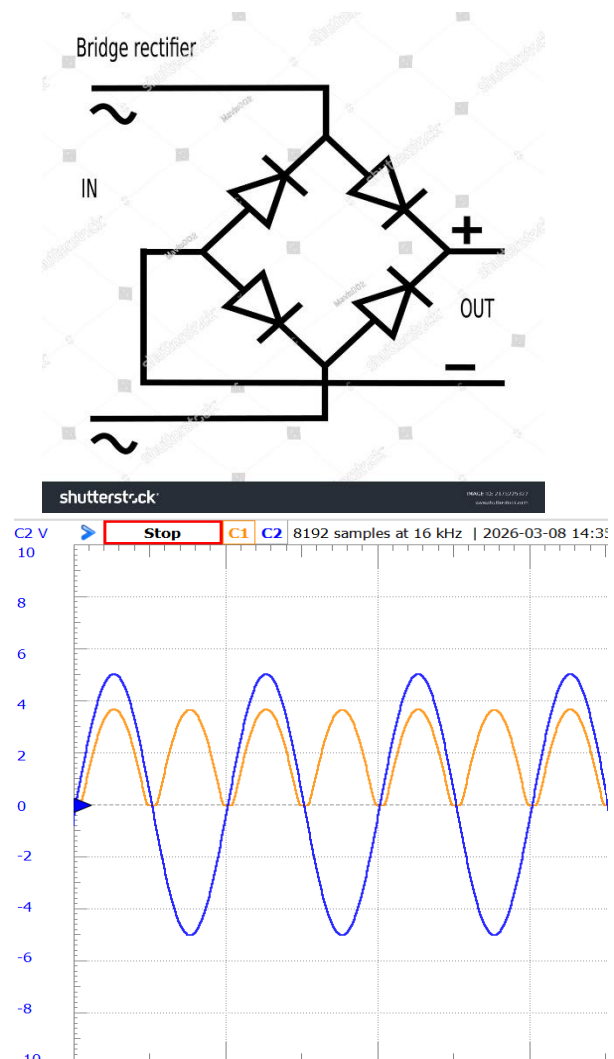
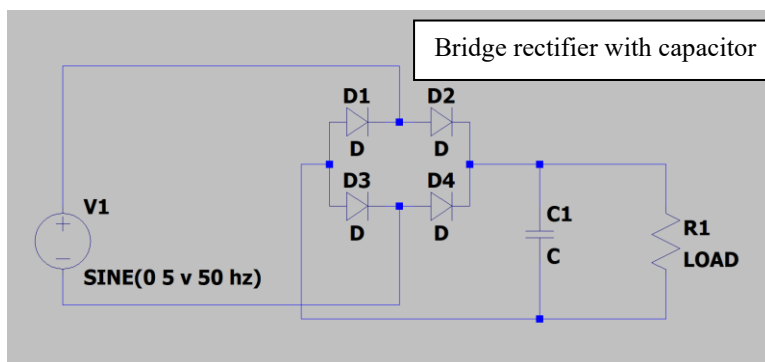


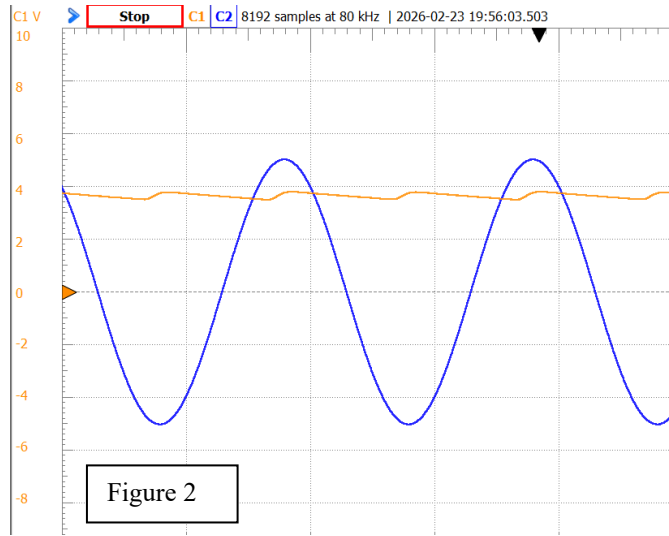
Figure 1.

Bridge Rectifier with Capacitor Filter:

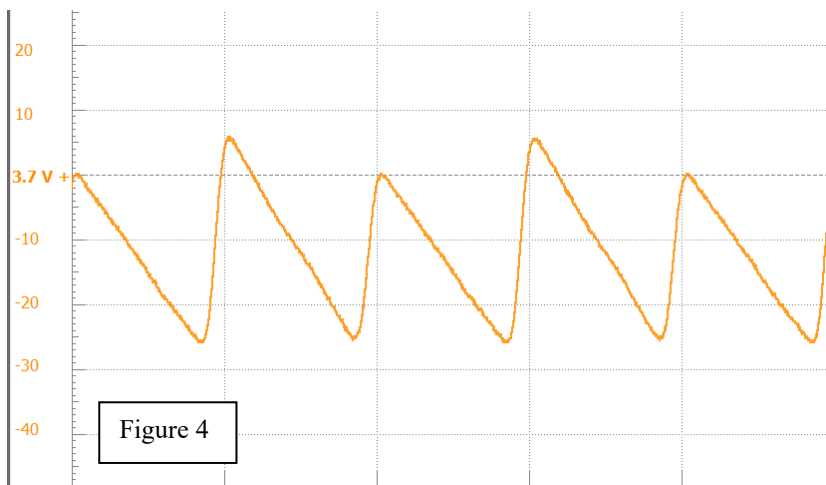
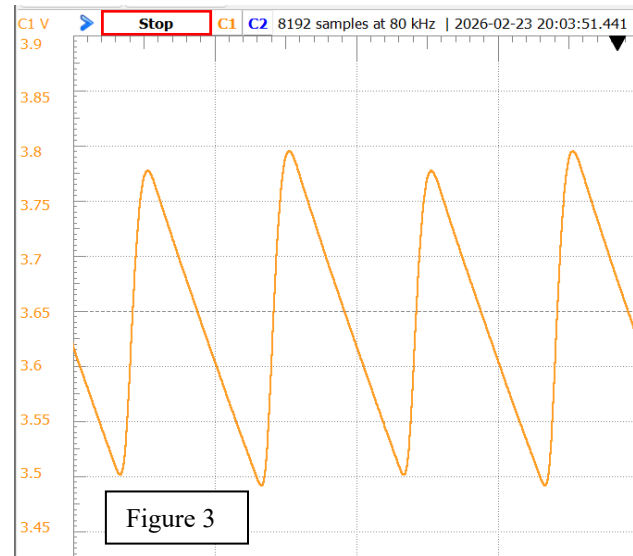


A capacitor added in parallel to the load serves to filter the wave by charging and discharging when V_{in} is negative. The result is a DC voltage with a ripple. The peak to peak range of the ripple can be attenuated by increasing the size of the capacitor.

Figure 2 and 3 depict the ripple when we use a $10\mu\text{F}$ capacitor. The ripples peak to peak fluctuation is 0.3 volts or 300 millivolts. Figure 3 depicts the ripple when we adjust the range on the oscilloscope. The scale in figure 3 is 50 millivolts per division. It's important to note that figures 2 and 3 depict the waves in both cases at the same frequency. Only the voltage axis is different between the two figures.



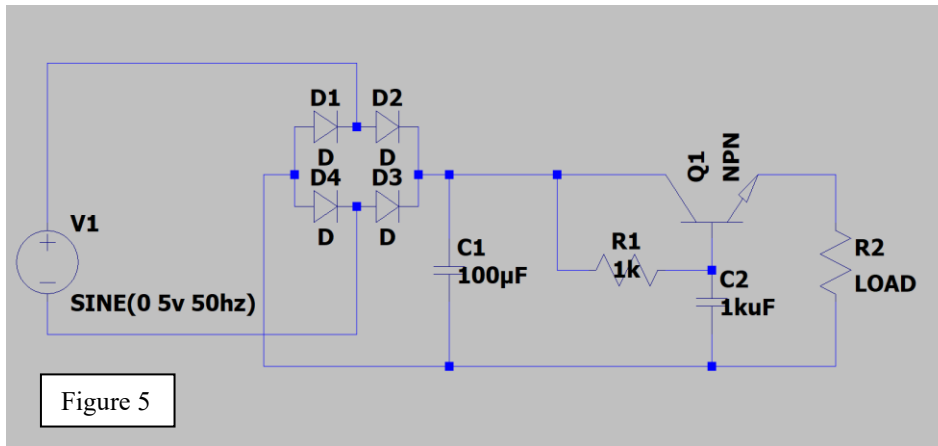
If we replace the $10\mu\text{F}$ capacitor with a $100\mu\text{F}$ one, the filtered signal is attenuated to about 30 millivolts. Below, figure 4 depicts the ripple. In this case, the voltage axis has been scaled to 10 millivolts per division. We can see that the peak-to-peak ripple is ~ 30 millivolts



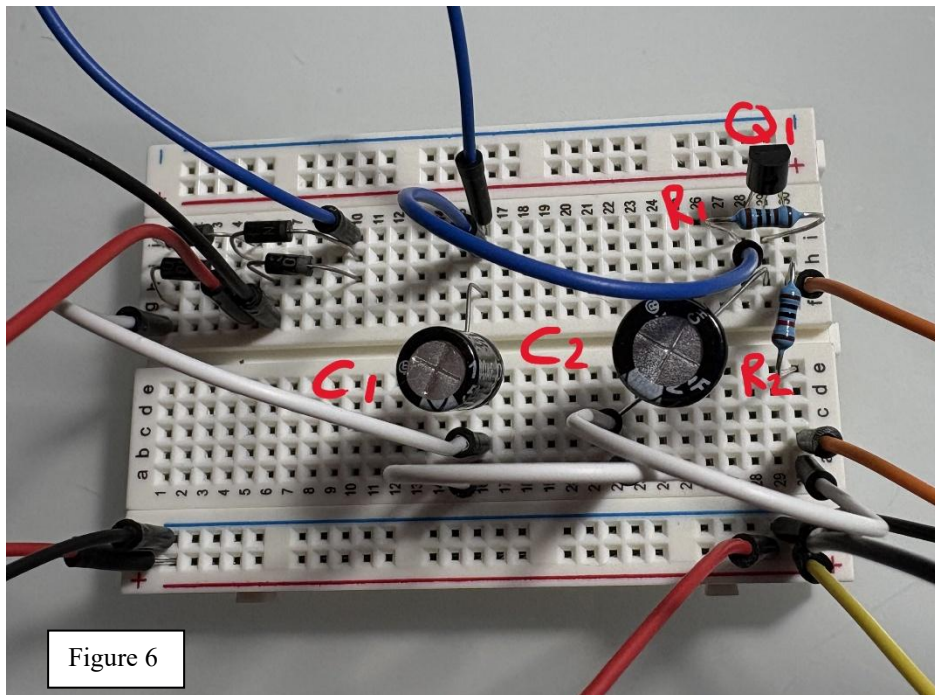
It's important to note too that in both of these cases, the voltage (after the diode network) has dropped from $5V_{max}$ to $\sim 3.6V$!

Bridge Rectifier with Capacitor Filter and Capacitance Multiplier:

If we take a NPN transistor and connect an RC network to the base and then connect it in series with our filtered voltage, we can artificially increase the capacitance in our circuit and bypass the voltage and current constraints. This essentially gives the circuit a capacitor that can charge and discharge much more charge than a capacitor in parallel with the load. Figure 5 and 6 depict the circuit and the breadboard prototype respectively.



The labeled components in Figure 6 correspond to the labeled circuit elements in Figure 5. On the breadboard, 4 1N4007 diodes were used in the diode network, a 100μF capacitor was used as the first filter, a 1k resistor, a 1kμF capacitor and NPN transistor were used for the capacitance multiplier. The load resistor is arbitrary.



The capacitance multiplier was able to attenuate the ripple to only 2 millivolts peak-to-peak! Figure 7 shows the resulting 'waveform' (at this point it's almost closer to a DC voltage). The voltage axis on the oscilloscope graph is calibrated to 2 millivolts per division.

One important note, however, one drawback of using the capacitance multiplier is that the voltage drops by about $0.6V$ across the transistor. The final voltage across the load is a $\sim 3.1V$ DC signal with only a 2 millivolt ripple.

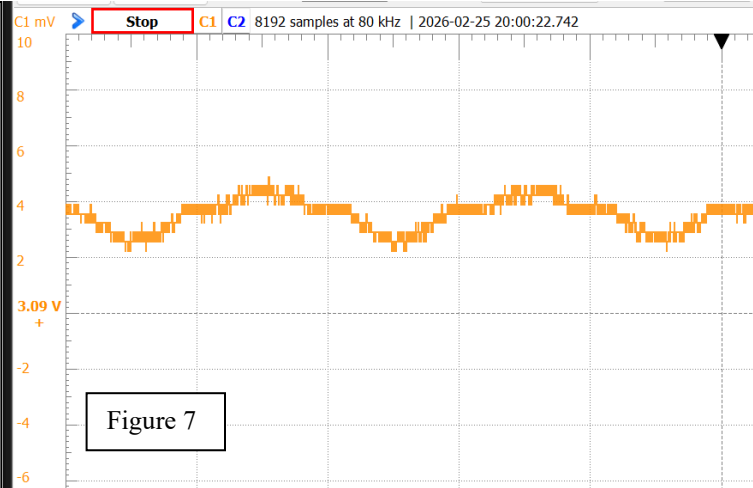


Figure 7

Parts used:

- Digilent Analog Discovery 2 (For waveform generation and oscilloscope)
- 4 X 1N4007 diodes
- $10\mu F$ electrolytic capacitor
- $100\mu F$ electrolytic capacitor
- $1k\mu F$ electrolytic capacitor
- $1k\Omega$ Resistor

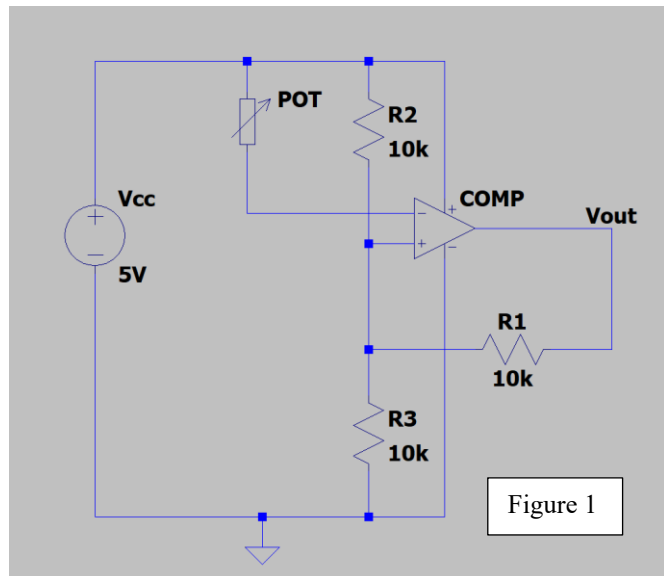
Schmitt Trigger

Context:

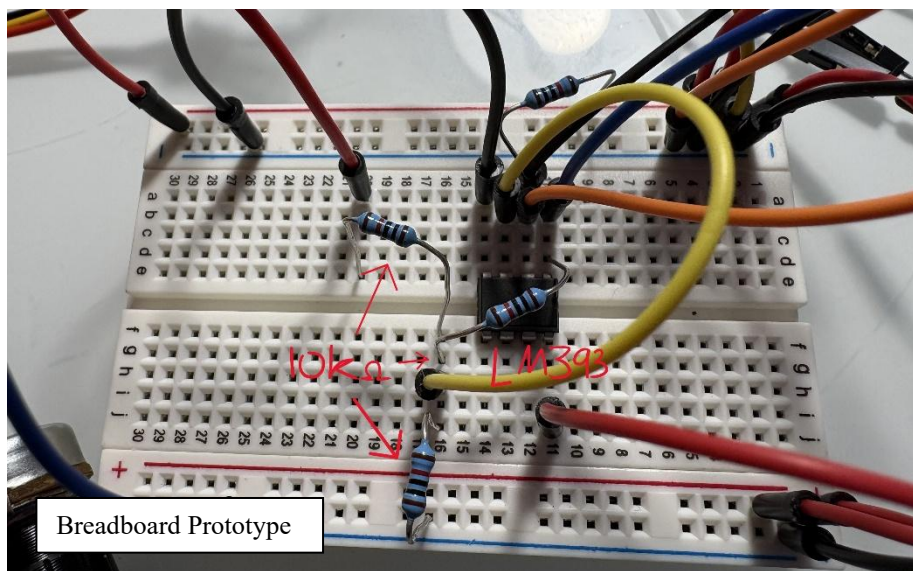
For this project, the goal is to design a circuit that maintains a steady HIGH or LOW signal despite rapid fluctuations or electrical noise. This is what the Schmitt Trigger excels at.

To be specific, the Schmitt Trigger in this portfolio entry is an inverting one with non-symmetrical thresholds. Using methods I'll outline below, we can actually choose where the threshold voltages are!

The thresholds of the trigger vary depending on the choice of resistors in the network. If V_{out} is 0V (the LOW state) the feedback resistor is parallel with the grounded resistor. Their voltage forms the LOW threshold. If V_{out} is 5V (the HIGH state) the feedback resistor is in parallel with the input resistor, thus their voltage is the upper threshold.

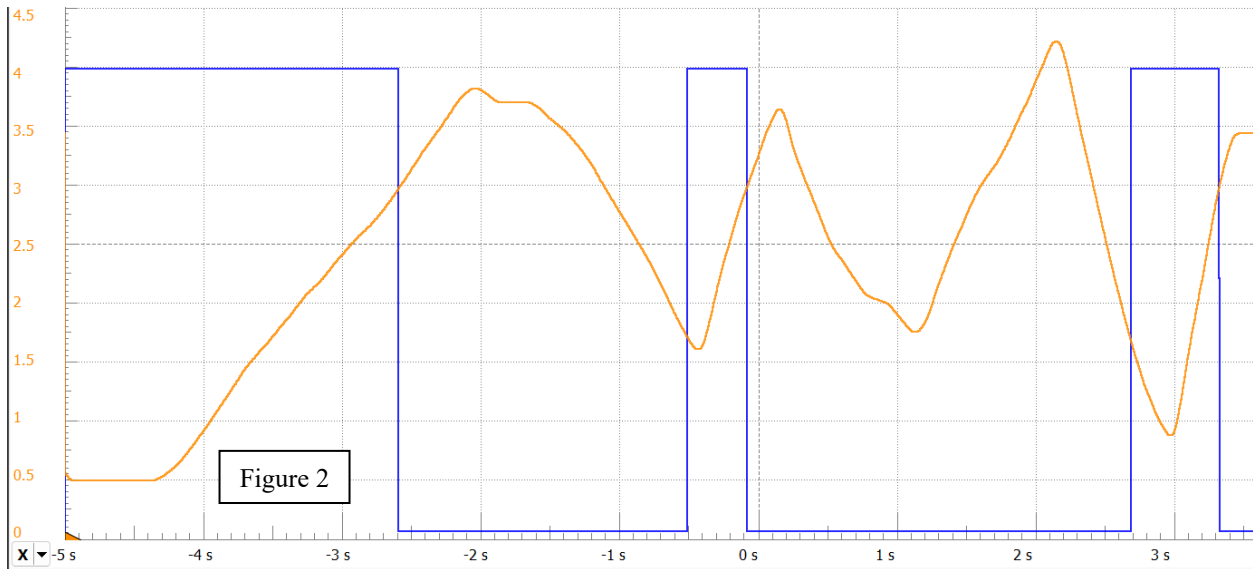


I constructed the circuit in Figure 1 with a potentiometer (10k Ω) to model an unpredictable input signal. 10k Ω resistors were used in the resistor network. Calculating the voltage drop across 2 parallel 10k Ω resistors in series with a singular 10k Ω resistor yields thresholds at (under ideal op amp circumstances) 3.3V and 1.66V (Hi and Lo)

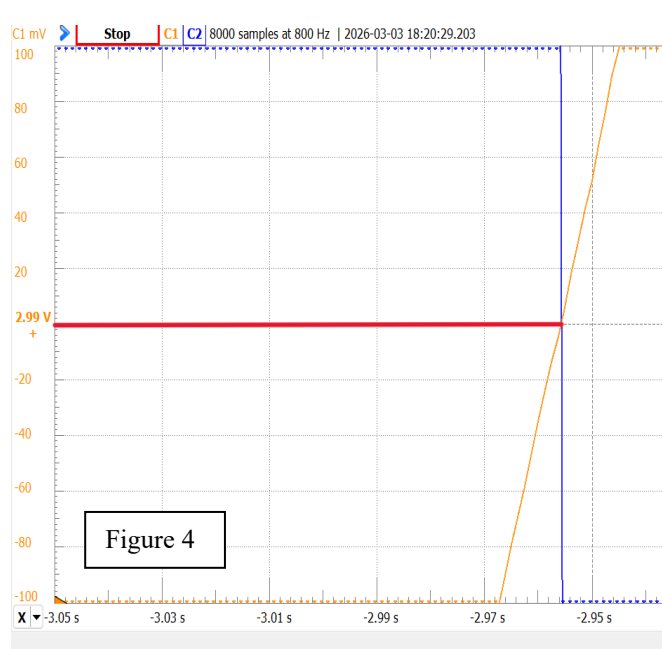
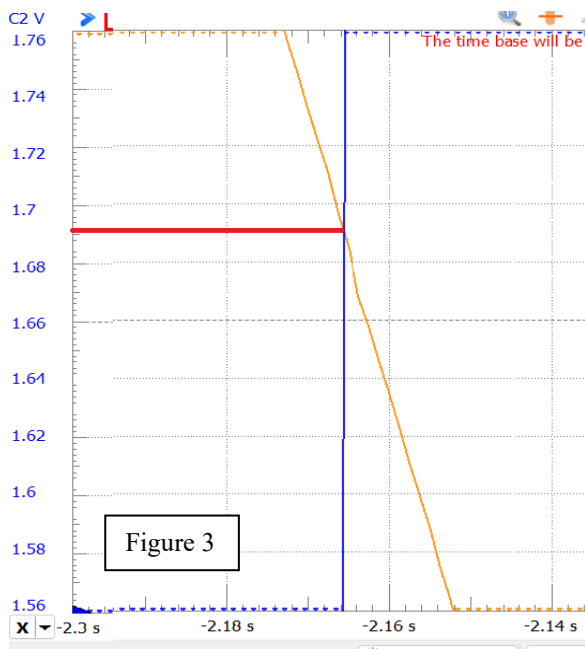


Oscilloscope Readings:

With the circuit built and receiving power from the ADK2, manipulated the potentiometer unpredictably as show in Figure 2. The orange line depicts the input voltage on the inverting lead of the comparator. The blue line corresponds with the output voltage.



Upon closer inspection of the rising and falling edge of the square waveform, we can see from Figures 3 and 4 that the intersection of V_{in} and V_{out} correspond to the voltages 1.69V and 2.99V respectively.



Integrator

Context:

The purpose of an integrator circuit is to take an input signal and integrate it with respect to time. In order to achieve this, an op amp can be used in lieu of a feed back resistor. I will call it the feedback capacitor. Figure 1 shows the circuit. Figure 2 shows the breadboard prototype.

In an op amp,

$$V_- = V_+ \text{ and } I_- = I_+ = 0$$

The input of the op amp, per Ohms law is

$$I = \frac{V_{in}}{R_{in}}$$

The current through a capacitor is

$$I = C \frac{dV_{out}}{dt}$$

In this case

$$\frac{V_{in}}{R_{in}} = C \frac{dV_{out}}{dt}$$

So

$$V_{out} = -\frac{1}{R_{in}C} * \int V_{in}(t)dt$$

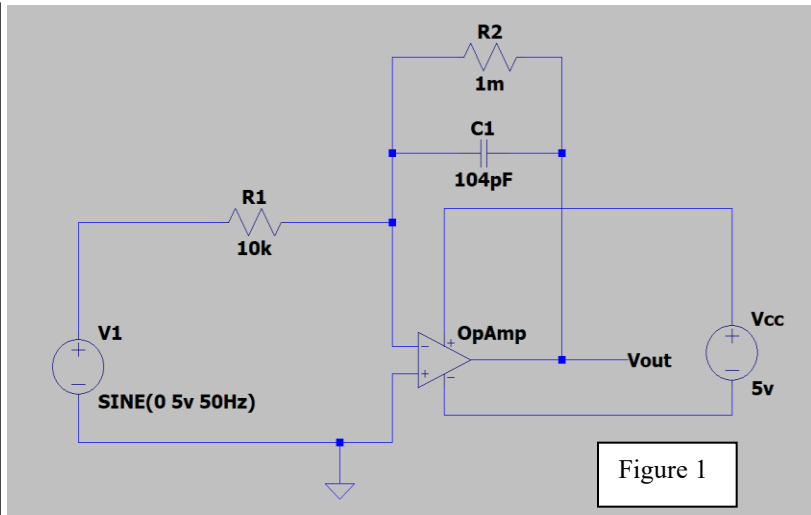


Figure 1

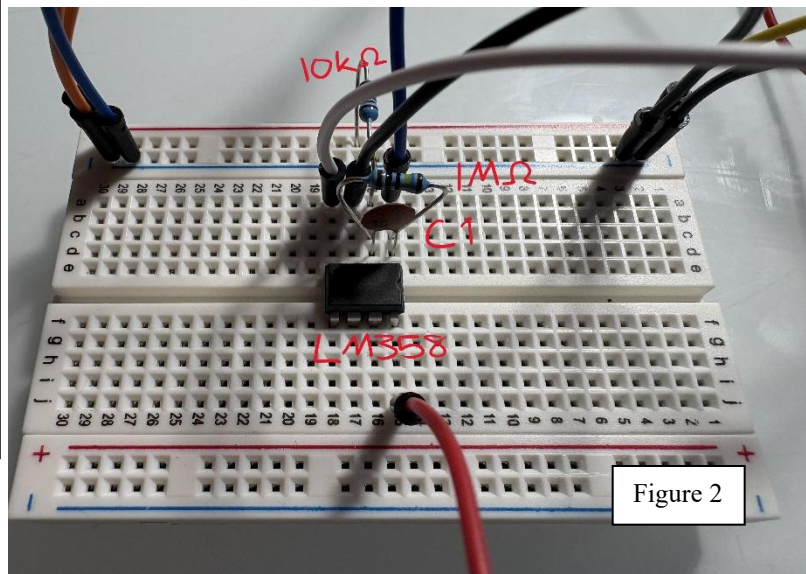


Figure 2

A smaller capacitor in this instance results in a waveform that is less attenuated. Larger capacitors would result in a more diminished integral wave. In this instance, a 104nF ceramic capacitor was used.

Oscilloscope readings:

It's important to remind the viewer that the input of the op amp is into the **inverting** input. It's for this reason that all of the oscilloscope graphs will show inverted integral waves. Another thing to note is that the op amp's gain will increase the signal sizes of several of the integrals. Op amp gain was not the focus of this portfolio entry. In all the graphs, the orange wave is the V_{in} and the blue wave is V_{out} .

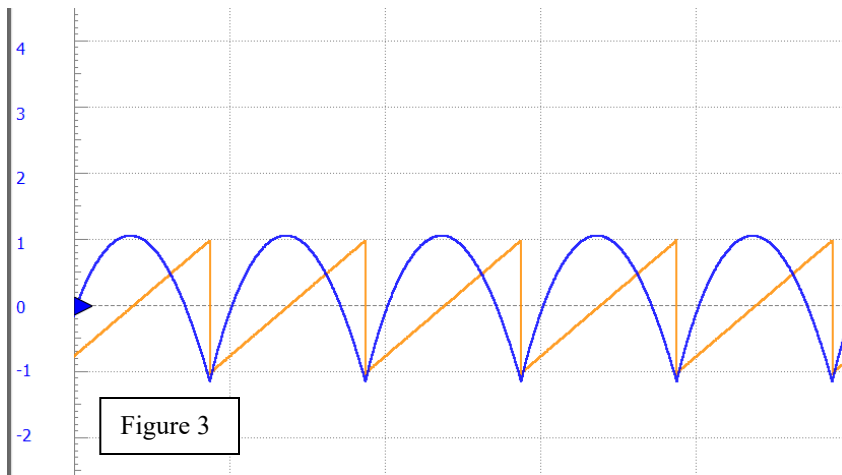


Figure 3 depicts a sawtooth input wave. Its integral is a strobing wave, much like the one seen in the bridge rectifier circuit.

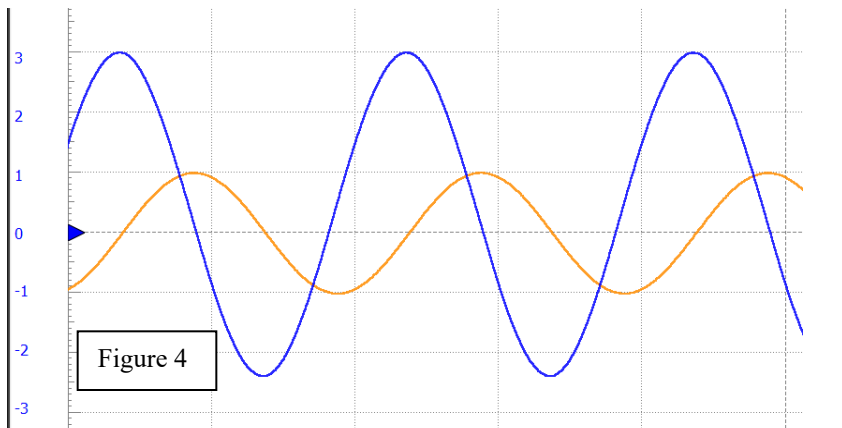
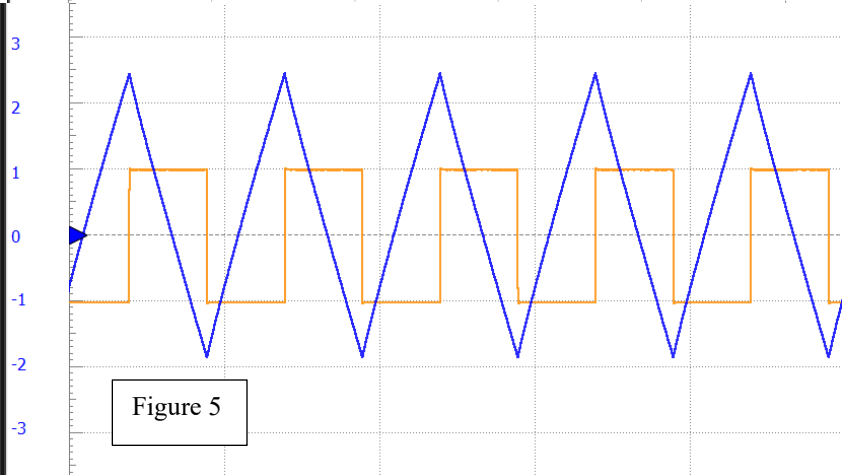


Figure 4 shows the graph with a sine wave input. We can see the inflection point of the output wave corresponds on the x axis with the peaks of the input wave.



The final wave in Figure 5 shows a square input wave. The resulting integral wave is triangular. Much like the sine wave, the 'inflection' point (even though there isn't actually an inflection point on a straight line like that) corresponds with the center of the square wave.

Half Adder

Context:

Half adders are one of the most basic elements of digital circuits. Their main purpose is to take two single digit binary numbers and add them together. Half adders are used extensively in ALUs in processors to perform calculations on numbers and instructions.

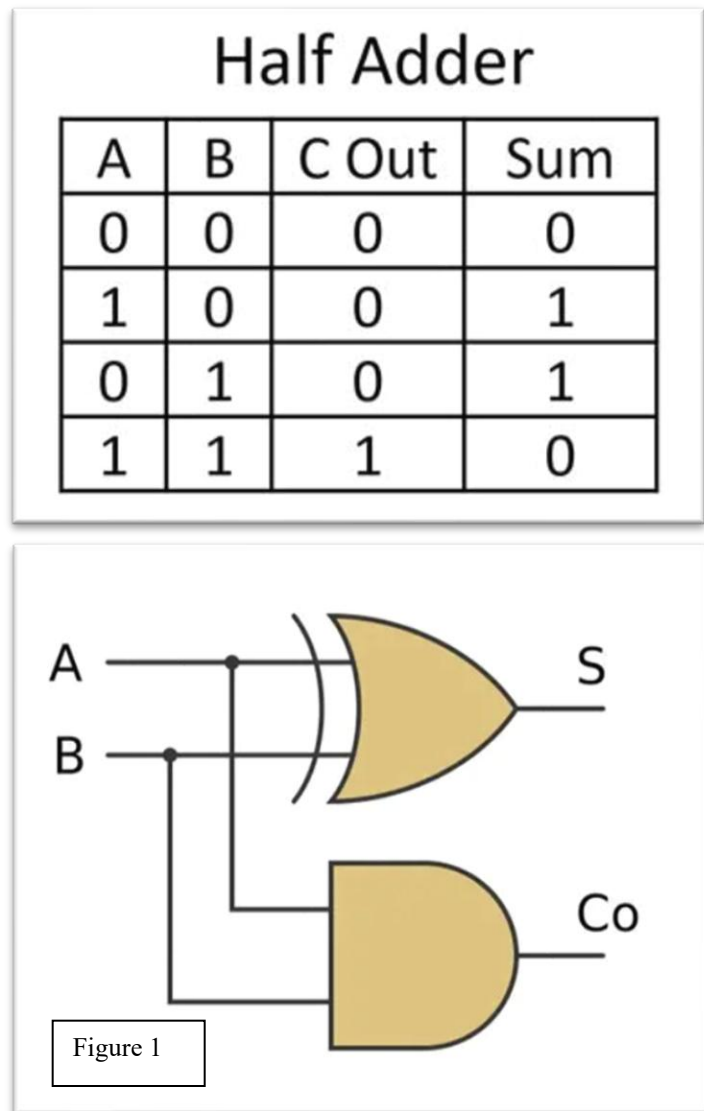
To the right is a truth table of a half adder. It has two inputs; A and B. If the inputs are not the same they will return a sum of 1, or HIGH. If they are both 0, they will return a sum and carry of 0. If they are both one, they will carry a 1.

The Boolean algebra of the half adder is as follows

$$A \oplus B = SUM$$

$$AB = CARRY$$

Figure 1 depicts the logic gates of these two operations.



The Digilent Analog Discovery 2 has 16 logic I/O pins that can generate 3.3V signals to be used by Logic gate ICs. My breadboard prototype circuit utilized 2 ICs: the 74LS86 and 74LS08. Both ICs require 5V DC to their power rails. Each pin has a HIGH threshold of 2 – 5V, so the 3.3V output signals on the ADK2 are sufficient.

Figures 2, 3 and 4 show the Waveforms software interface that controls the outputs on the ADK2. They can be set to mimic switches, pushbuttons, analog sliders, etc. For this portfolio entry I am using single throw switches. The switches being used are 14 and 15.

Figures 2 and 3 show when only one input is HIGH. These two settings correspond with Figure 5 which depicts the red LED illuminated on the breadboard circuit. Figure 4 shows both inputs set to HIGH and corresponds with the blue LED illuminated in figure 6.

